Jiachuan Wang

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SUMMARY:

PhD candidate in Computational Neuroscience at the National University of Singapore developing biologically plausible models of associative memory and spatial navigation. My research integrates computational modeling, machine learning, and neuroimaging to investigate learning, memory, and decision-making across biological and artificial systems.

EDUCATION:

National University of Singapore — Singapore
Ph.D. candidate in Medicine — *Aug. 2023 – Present*
Neuroscience Track and Biostatistics, Bioinformatics & Epidemiology Track

Zhejiang University (Joint Degree Programme) — Hangzhou, China
B.S. in Bioinformatics — *Sep. 2019 – Jun. 2023*

The University of Edinburgh — Edinburgh, UK
B.S. (Hons) in Biomedical Informatics — *Sep. 2019 – May. 2023*

EXPERIENCE:

The N.1 Institute for Health, National University of Singapore — Singapore
Graduate Researcher — *Sep. 2023 – Present*
Advisors: Andrew Tan, Camilo Libedinsky, Shih-Cheng Yen

- Developed biologically plausible reinforcement learning models based on the spike response model, enabling successful learning of multiple paired-association navigation tasks.
- Developed a representation learning model incorporating Hebbian plasticity and neurogenesis that reproduces experimentally observed pattern separation effects not captured by canonical reinforcement learning models.

Centre for Discovery Brain Sciences, The University of Edinburgh — Edinburgh, UK
Research Intern; Advisor: Gediminas Lukšys — *Mar. 2022 – Aug. 2023*

- Compared emotion and memory decoding performance across real fMRI datasets and a meta-analysis database using multiple machine learning models, providing evidence for systematic biases associated with commonly used statistical correction procedures in neuroimaging.
- Estimated parameters of biologically inspired reinforcement learning models incorporating place, distance, and border cells, identifying computational correlates underlying temperature-based behavioral differences.

School of Brain Science and Brain Medicine, Zhejiang University — Hangzhou, China
Research Intern; Advisor: Zhiping Wang — *Jan. 2022 – Sep. 2022*
Jun. 2020 – Aug. 2020

- Analyzed label-free quantification proteomics datasets and conducted downstream enrichment analyses.

Interdisciplinary Institute of Neuroscience and Technology, Zhejiang University Hangzhou, China

Research Intern; Advisor: Anna Wang Roe *Apr. 2021 – Oct. 2021*

- Developed a web-based fMRI data visualization platform featuring automated anatomical labeling function to facilitate data exploration and interpretation.

PUBLICATIONS:

• Kong, X., Shu, X., **Wang, J.**, Liu, D., Ni, Y., Zhao, W., Wang, L., Gao, Z., Chen, J., Yang, B., Guo, X. and Wang, Z. (2022) Fine-tuning of mTOR signaling by the UBE4B-KLHL22 E3 ubiquitin ligase cascade in brain development. ***Development***. doi: 10.1242/dev.201286.

• Zhang, L., Ma, X., Wu, Z., Liu, J., Gu, C., Zhu, Z., **Wang, J.**, Shu, W., Li, K., Hu, J. and Lv, X. (2022) Prevalence of ground glass nodules in preschool children: a cross-sectional study. ***Translational Pediatrics***. doi: 10.21037/tp-22-465.

MANUSCRIPTS:

• Shetru, V. J.*, **Wang, J.***, Kumar, M. G., Xiong, V. T., Libedinsky, C., Yen, S.-C., Tan, A. Y.-Y. and Polepalli, J. S. Cerebellin-4 suppresses activity-dependent hippocampal neurogenesis and promotes pattern separation memory. Under review at *Communications Biology*.

• Koh, J. H., Kalimuthu, B., **Wang, J.**, Xiong, V. T., Shetru, V. J., Tan, A. Y.-Y. and Polepalli, J. S. Cerebellin-4 organizes hippocampal inhibitory long-term potentiation. Under review at *PNAS*.

• Wang, S.*, **Wang, J.***, Zhu, W., Cheng, Y., Aydemir, B., Qiu, Y., Gerstner, W., Sandi, C. and Luksys, G. Computational model-based analysis: Modeling effects of stress, strain and task difficulty on mice spatial navigation strategies in Morris Water Maze. In Preparation.

INVITED TALKS:

• Singapore Neuroscience Association Symposium *Jun. 2026*

• Neurobiology Programme, National University of Singapore *Apr. 2025*

SELECTED CONFERENCE PRESENTATIONS:

• Koh, J. H., Kalimuthu, B., **Wang, J.**, Xiong, V. T., Goh, L. Z.-N., Tan, A. Y.-Y. and Polepalli, J. S. Cerebellin-4 instructs hippocampal inhibitory LTP required for contextual memory precision. Poster presentation, **Singapore Neuroscience Association** Symposium, Singapore, June 2026.

• **Wang, J.**, Shetru, V. J., Kumar, M. G., Libedinsky, C., Yen, S.-C., Tan, A. Y.-Y. and Polepalli, J. S. Unsupervised Hebbian learning drives biologically interpretable pattern separation in a hippocampal–striatal network. Spotlight talk and poster presentations, **AAAI Workshop** on Neuro for AI & AI for Neuro, Singapore, January 2026. doi: 10.48448/jsed-nf31.

• Wang, S., **Wang, J.**, Zhu, W., Cheng, Y., Aydemir, B., Qiu, Y., Gerstner, W., Sandi, C. and Luksys, G. Computational model-based analysis of spatial navigation strategies under stress and uncertainty using place, distance and border cells. Poster presentation, **Society for Neuroscience (SfN)** Meeting, Washington, D.C., November 2023.

GRANTS & AWARDS:

NUS Research Scholarship
NUS Graduate School — *2023 – 2028*

Outstanding Graduate of Zhejiang University
Zhejiang University — *2023*

Zhejiang University Scholarship
Zhejiang University — *2022*

Academic Scholarship (¥40,000)
ZJU-UoE Institute — *2022*

Student Research Training Program Funding (¥1,200)
Zhejiang University — *2021 – 2022*

TEACHING:

National University of Singapore
Graduate Teaching Assistant

- Systems Neurobiology — *Fall 2024, Fall 2025*
- Beginning Artificial Intelligence Through Neuroscience — *Fall 2024*

REVIEWING:

- Journal: *Scientific Reports*.
- Conference: Computational and Systems Neuroscience (COSYNE), Cognitive Computational Neuroscience (CCN).

ASSOCIATIONS:

- Society for Neuroscience, Singapore Chapter
- ALBA Network

SKILLS:

- **Languages**: Mandarin (native), English (fluent).
- **Programming Languages**: Python, R, PostgreSQL, Bash. Basics of: C/C++, Java.
- **Machine Learning & Data Analysis**: scikit-learn, Nilearn, Keras, AFNI.
- **Applications**: Git, LaTeX.